

Project Design Phase

Problem – Solution Fit Template

Date	26 JUNE 2026
Project Name	Metro Ticket Booking System
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Metro Ticket Booking System is designed to solve the challenges faced by metro passengers during ticket booking. The system enables users to book tickets online, automatically calculates fares based on source and destination stations, generates secure QR code tickets, and stores booking details digitally. This solution reduces waiting time, eliminates paper tickets, and provides a fast, secure, and contactless metro ticket booking experience.

Purpose:

- ☐ Provide an online metro ticket booking system.
- ☐ Reduce waiting time at metro ticket counters.
- ☐ Automatically calculate ticket fares.
- ☐ Generate secure QR-code based digital tickets.
- ☐ Improve passenger convenience using ServiceNow
- ☐ Promote paperless and contactless travel.

Template:

Define CS, fit into CC	1. CUSTOMER SEGMENT(S) CS	6. CUSTOMER CONSTRAINTS CC	5. AVAILABLE SOLUTIONS AS	Explore AS, differentiate	
	- Daily Metro Commuters - College Students - Office Employees - Tourists - Senior Citizens - Metro Railway Administration	- Time-consuming booking - Long waiting time - Paper ticket damage/loss - Manual fare errors - Peak-hour congestion	- Manual ticket counters - Paper tickets - Metro smart cards - Existing metro mobile apps - Online payment systems		
Focus on J&P, tap into BE, understand RC	2. JOBS-TO-BE-DONE / PROBLEMS J&P	9. PROBLEM ROOT CAUSE RC	7. BEHAVIOUR BE		Focus on PIR, tap into BE, understand RC
	- Book metro tickets quickly. - Avoid waiting in long queues. - Calculate ticket fare automatically. - Generate digital QR tickets. - Store booking details securely. - Enable paperless ticketing.	- Manual ticket booking process - Lack of automation - No QR-based ticketing - Slow ticket generation - Heavy passenger traffic	Online - Uses smartphone - Prefers online booking - Makes digital payments - Scans QR code Offline - Visits ticket counters - Uses metro smart card - Requests printed tickets		
Identify strong TR & EM	3. TRIGGERS TR	10. YOUR SOLUTION SL		8. CHANNELS of BEHAVIOUR CH	Extract online & offline CH of BE
	- Long queues at ticket counters. - Growing use of smartphones. - Demand for contactless travel. - Digital payment adoption. - Smart city initiatives.	Develop a Metro Ticket Booking System using ServiceNow that allows passengers to book tickets online, automatically calculate fare based on source and destination, generate a dynamic QR Code before booking, securely store booking information in the Metro Ticket Booking table, and provide a fast, paperless, contactless travel experience.		ONLINE - ServiceNow Service Portal - Mobile Browser - Metro Website - QR Scanner OFFLINE - Metro Stations - Ticket Counters - Self-Service Kiosks - Customer Help Desk	
4. EMOTIONS: BEFORE / AFTER EM					
Before - Waiting in queues - Paper ticket loss - Manual fare calculation - Booking delays After - Instant booking - QR-based ticket - Faster metro entry - Hassle-free travel					